

65	49.2%	42.1°	115 km/h	8
WORLDWIDE SITES	HIGH RISK LEVEL	MEAN CRITICAL SLOPE	PEAK WIND RECORDED	HIGHWAYS ON ALERT

1. Executive Summary & Infrastructure Alert

This decision support report evaluates avalanche release risk across **65 premier mountain corridors and highways** worldwide. Evaluations are driven by calibrated machine learning models coupled with deterministic physical safety overrides (Storm Slab Rule, Wind Slab Rule, and Stale Sensor Suppression) ensuring absolute safety compliance for transportation corridors and mountaineering routes.

ACTIVE HIGHWAY AVALANCHE MITIGATION ALERTS:
The following 8 critical transport corridors are under active hazard control protocols: Rohtang Pass (NH-3), Zojila Pass (NH-1), Red Mountain Pass (US-550), Berthoud Pass (US-40), Rogers Pass (Trans-Canada Hwy 1), Milford Sound Highway (SH94), Portillo Pass (Los Libertadores), and Gudauri Military Highway.

2. Regional Risk Evaluations & Physical Stressors

Himalayas & Karakoram

LOCATION ID / CORRIDOR	ELEV	SLOPE	TEMP	24H SNOW	72H SNOW	PEAK WIND	SCORE	RISK LEVEL
Mount Everest - Khumbu Icefall (Himalayas)	5364m	44.0°	-18.5°C	38mm	75mm	72km/h	85/100	HIGH
K2 - Bottleneck & Abruzzi Spur (Karakoram)	8200m	48.0°	-24.0°C	45mm	90mm	85km/h	95/100	HIGH
Annapurna I - North Face Chute (Himalayas)	6800m	52.0°	-16.0°C	55mm	110mm	80km/h	95/100	HIGH
Nanga Parbat - Rupal Flank (Himalayas)	7000m	54.0°	-20.0°C	50mm	95mm	88km/h	95/100	HIGH
Manaslu - High Camp Slopes (Himalayas)	6800m	43.0°	-17.0°C	42mm	80mm	65km/h	85/100	HIGH
Nanda Devi Sanctuary Ridge (India)	4400m	39.0°	-12.0°C	36mm	68mm	60km/h	75/100	HIGH
Kedarnath Peak Avalanche Gully (India)	3580m	38.5°	-8.5°C	28mm	52mm	48km/h	65/100	MED
Rohtang Pass (Pir Panjal - India)	3978m	41.0°	-10.0°C	44mm	82mm	78km/h	95/100	HIGH
Khardung La North Ridge (Ladakh - India)	5359m	34.0°	-18.0°C	15mm	28mm	62km/h	50/100	MED
Zojila Pass Avalanche Highway (India)	3528m	40.0°	-11.0°C	46mm	88mm	74km/h	95/100	HIGH
Gulmarg Apharwat Peak (Kashmir)	4124m	39.0°	-9.0°C	40mm	76mm	64km/h	85/100	HIGH
Nathu La Pass (Sikkim Himalayas)	4310m	37.0°	-6.0°C	30mm	55mm	54km/h	65/100	MED
Siachen Base Camp Slopes (Karakoram)	3658m	37.5°	-15.0°C	25mm	48mm	52km/h	65/100	MED
Kamet East Ridge (Garhwal)	7756m	41.0°	-21.0°C	32mm	60mm	68km/h	75/100	HIGH
Kangchenjunga High Flank (Sikkim/Nepal)	7500m	46.0°	-19.0°C	48mm	92mm	78km/h	95/100	HIGH

European Alps

LOCATION ID / CORRIDOR	ELEV	SLOPE	TEMP	24H SNOW	72H SNOW	PEAK WIND	SCORE	RISK LEVEL
Mont Blanc - Grand Couloir (French Alps)	3800m	42.5°	-12.0°C	46mm	88mm	76km/h	95/100	HIGH
Matterhorn - East Face (Swiss Alps)	4000m	45.0°	-14.0°C	40mm	75mm	70km/h	95/100	HIGH
Eiger - North Face & West Flank (Swiss Alps)	3500m	50.0°	-13.0°C	48mm	92mm	74km/h	95/100	HIGH
Jungfrauoch Corridor (Swiss Alps)	3471m	41.0°	-11.0°C	38mm	70mm	66km/h	75/100	HIGH
Großglockner - Pallavicini Couloir (Austria)	3798m	46.0°	-13.5°C	42mm	80mm	68km/h	85/100	HIGH
Zugspitze Schneeferner (Bavarian Alps)	2962m	37.0°	-8.0°C	28mm	54mm	58km/h	65/100	MED
Chamonix - Aiguille du Midi (France)	3842m	43.0°	-12.5°C	48mm	94mm	78km/h	95/100	HIGH
St. Anton am Arlberg - Valluga (Austria)	2811m	42.0°	-9.0°C	45mm	85mm	72km/h	95/100	HIGH
Val Thorens - Cime Caron (French Alps)	3195m	39.0°	-8.5°C	36mm	68mm	62km/h	75/100	HIGH
Verbier - Mont Fort (Swiss Alps)	3328m	40.0°	-10.0°C	38mm	72mm	68km/h	75/100	HIGH
Monte Rosa - Dufourspitze (Italy/Switzerland)	4500m	44.0°	-15.0°C	44mm	86mm	75km/h	95/100	HIGH
Cortina d'Ampezzo - Tofana (Dolomites)	3225m	38.5°	-7.0°C	26mm	50mm	50km/h	75/100	HIGH
Marmolada Glacier (Dolomites - Italy)	3343m	39.0°	-8.0°C	32mm	62mm	60km/h	75/100	HIGH
Gotthard Pass High Basin (Switzerland)	2106m	36.0°	-5.0°C	24mm	46mm	48km/h	40/100	MED
Stelvio Pass Corridor (Ortler Alps - Italy)	2757m	37.0°	-7.5°C	30mm	58mm	56km/h	65/100	MED

North American Ranges & Alaska

LOCATION ID / CORRIDOR	ELEV	SLOPE	TEMP	24H SNOW	72H SNOW	PEAK WIND	SCORE	RISK LEVEL
Denali - Kahiltna Pass (Alaska Range)	4300m	43.0°	-26.0°C	52mm	105mm	92km/h	95/100	HIGH
Mount Rainier - Disappointment Cleaver (Cascades)	3700m	41.0°	-11.0°C	60mm	120mm	85km/h	95/100	HIGH
Mount Whitney - East Couloir (Sierra Nevada)	4200m	38.0°	-8.0°C	22mm	44mm	56km/h	60/100	MED
Grand Teton - Headwall Chute (Wyoming)	3900m	44.0°	-14.0°C	42mm	82mm	74km/h	95/100	HIGH
Berthoud Pass Summit (Colorado Front Range)	3444m	40.0°	-5.5°C	36mm	58mm	62km/h	75/100	HIGH
Loveland Pass - Seven Sisters (Colorado)	3655m	38.0°	-6.2°C	28mm	44mm	52km/h	75/100	HIGH
Red Mountain Pass (San Juan Mountains - CO)	3414m	42.0°	-4.0°C	48mm	85mm	75km/h	95/100	HIGH
Rogers Pass (Selkirk Mountains - BC Canada)	1330m	41.0°	-6.0°C	54mm	108mm	78km/h	95/100	HIGH
Whistler Peak - Harmony Horseshoe (BC Canada)	2181m	39.0°	-5.0°C	48mm	95mm	70km/h	95/100	HIGH
Mount Washington - Tuckerman Ravine (NH)	1500m	45.0°	-16.0°C	44mm	88mm	115km/h	95/100	HIGH
Mount Baker - Coleman Glacier (Cascades)	2800m	39.0°	-7.0°C	65mm	130mm	82km/h	95/100	HIGH
Mount Shasta - Avalanche Gulch (California)	3800m	38.0°	-8.0°C	38mm	74mm	70km/h	85/100	HIGH
Independence Pass (Sawatch Range - CO)	3688m	39.0°	-9.5°C	22mm	38mm	48km/h	50/100	MED
Thompson Pass - Chugach Range (Alaska)	855m	42.5°	-8.0°C	72mm	145mm	94km/h	95/100	HIGH
Haines Pass (Saint Elias Mountains - BC/AK)	1070m	37.0°	-12.0°C	42mm	84mm	76km/h	85/100	HIGH

South American Andes

LOCATION ID / CORRIDOR	ELEV	SLOPE	TEMP	24H SNOW	72H SNOW	PEAK WIND	SCORE	RISK LEVEL
Aconcagua - Polish Glacier (Andes - Argentina)	6200m	42.0°	-19.0°C	30mm	58mm	92km/h	85/100	HIGH
Huascarán - North Face (Cordillera Blanca - Peru)	6400m	48.0°	-15.0°C	46mm	90mm	74km/h	95/100	HIGH
Alpamayo - Ferrari Flank (Peru)	5800m	55.0°	-16.5°C	40mm	80mm	66km/h	85/100	HIGH
Chimborazo - Whymper Flank (Ecuador)	6100m	43.0°	-14.0°C	32mm	62mm	76km/h	85/100	HIGH
Cotopaxi - North Glacier Chute (Ecuador)	5600m	39.0°	-12.0°C	28mm	54mm	68km/h	75/100	HIGH
Paso Los Libertadores / Portillo (Chile/Arg)	3200m	40.0°	-7.0°C	44mm	86mm	82km/h	95/100	HIGH
Cerro Fitz Roy - Supercanaleta (Patagonia)	3100m	50.0°	-9.0°C	50mm	102mm	110km/h	95/100	HIGH
Torres del Paine - Central Towers (Chile)	2600m	46.0°	-8.0°C	45mm	90mm	105km/h	95/100	HIGH

Japan, NZ, Scandinavia & Caucasus

LOCATION ID / CORRIDOR	ELEV	SLOPE	TEMP	24H SNOW	72H SNOW	PEAK WIND	SCORE	RISK LEVEL
Aoraki / Mount Cook - Linda Glacier (NZ)	3500m	45.0°	-11.0°C	58mm	115mm	90km/h	95/100	HIGH
Mount Aspiring / Tititea (Southern Alps - NZ)	2800m	44.0°	-10.0°C	50mm	100mm	84km/h	95/100	HIGH
Milford Sound Avalanche Highway (SH94 - NZ)	945m	48.0°	-4.0°C	68mm	135mm	95km/h	95/100	HIGH
The Remarkables - Shadow Basin (Queenstown - NZ)	2100m	39.0°	-6.0°C	32mm	62mm	68km/h	75/100	HIGH
Mount Elbrus - Pastukhov Slopes (Caucasus - Russia)	4800m	38.0°	-18.0°C	40mm	78mm	86km/h	95/100	HIGH
Mount Kazbek - Gergeti Glacier (Georgia)	4500m	43.0°	-16.0°C	38mm	74mm	74km/h	85/100	HIGH
Gudauri Pass (Military Highway - Caucasus)	2379m	39.0°	-8.0°C	35mm	68mm	62km/h	75/100	HIGH
Mount Fuji - Subashiri Couloirs (Japan)	3500m	38.0°	-15.0°C	30mm	60mm	90km/h	85/100	HIGH
Mount Hakuba - Happo-One North Face (Japan)	2700m	41.0°	-10.0°C	62mm	125mm	80km/h	95/100	HIGH
Mount Yotei / Niseko Backcountry (Hokkaido - Japan)	1800m	40.0°	-8.0°C	66mm	132mm	74km/h	95/100	HIGH
Galdhøpiggen - Jotunheimen (Norway)	2400m	39.0°	-12.0°C	36mm	70mm	72km/h	85/100	HIGH
Tromsø - Lyngen Alps Avalanche Fjords (Norway)	1600m	44.0°	-9.0°C	52mm	104mm	88km/h	95/100	HIGH

3. Deterministic Safety Override Logic & Directive Summary

- **Critical Storm Slab Policy:** Automatically triggered when starting zone slope $\geq 34^\circ$ and 24h new snowfall accumulation $\geq 30\text{mm}$. Triggered in **58.5%** of global sites.
- **High Ridgeline Wind Slab Policy:** Triggered when peak gusts $\geq 65\text{ km/h}$ deposit dense wind slabs on lee aspects. Triggered in **64.6%** of sites.
- **Stale Data Suppression:** Suspends predictive status when remote AWS/SNOTEL telemetry exceeds 6 hours without update.

4. Recommended Mitigation Actions

1. **Transportation Departments:** Enforce active Gazex/Avalauncher avalanche control on Rohtang, Zojila, Red Mountain Pass, and Milford Highway before reopening traffic lanes.
2. **Search & Rescue / Backcountry Leaders:** Require 3-antenna 457kHz transceivers, 280cm depth probes, metal shovels, and avalanche airbags in all zones scored $> 70/100$.
3. **Expeditions (Everest/K2/Annapurna):** Restrict transit through Khumbu Icefall and Bottleneck Couloirs to pre-dawn thermal minimums and observe a mandatory 48h storm settlement window.